Midterm Exam for CaSB 186/286
April 25, 2018

1. **(20 points)** In your homework we studied a problem with super-exponential growth. One form of super-exponential growth is given by population growth rate, $dN(t)/dt$, being proportional to the 3rd power of the population size, $N^3(t)$.
a. Write down the continuous differential equation that corresponds to the dynamics described above.
b. Write down the discrete version of the equation from part a. Express your result in terms of how the population size, $N(t+1)$, at time $t+1$ equals (depends on) involving the population size, $N(t)$, at time t.
c. Iterate once the equation in part b. by writing down the equation for $N(t)$ in terms of $N(t\text{-}1)$ and then substitute this into your equation from part b. In looking at your new version of this equation, explain whether you think it will be easy or hard to find a simple solution for this discrete equation?
d. Solve the continuous part of the equation from part a. To solve, follow the same procedure as you did for your homework.
f. When $N(0)=2$, write down an expression for the time at which the population size would be predicted to go to infinity. If there are constants for which you have not been given values, you do not have to calculate an actual number for this time, and you can just leave your answer in terms of symbols.

2. **(20 points)** Consider a system in which Y_1 represents the concentration of one protein and Y_2 represents the concentration of a different protein. When a protein of type Y_1 interacts with a protein of type Y_2, the interaction converts the protein of type Y_1 into a protein of type Y_2 while the protein of type Y_2 remains unchanged. This situation approximately describes how prion diseases spread. Prions are responsible for brain diseases such as "mad-cow" disease. We can begin to describe the dynamics of this system in analogy to predator-prey or Lotka-Volterra equations where the protein of type Y_2 is like the predator and the protein of type Y_1 is like the prey. The corresponding equations would thus be.

$$\frac{\mathrm{d}Y_1}{dt} = rY_1 - \bar{C}Y_1Y_2$$
$$\frac{\mathrm{d}Y_2}{dt} = \varepsilon\bar{C}Y_1Y_2 - ZY_2$$

Explain what the parameters r, $\bar{C}$, ε, and Z mean in the context of this model for prions. Are there any of these parameters that are no longer needed for this simple model? Explain why each term depends on Y_1 and Y_2 in the way it does, including the overall sign of the term.

3. **(30 points)**The standard Lotka-Volterra equations for predator-prey systems are

$$\frac{\mathrm{d}N}{dt} = rN - \bar{C}NP$$
$$\frac{\mathrm{d}P}{dt} = \varepsilon\bar{C}NP - ZP$$

where r is population growth rate for the prey, $\bar{C}$ is the consumption rate for the predator population on the prey population, ε is the efficiency of converting prey biomass into predator biomass, Z is the intrinsic mortality rate for the predator

species, N is the number density of the prey species, and P is the number density of the predator species.

a. In these equations, no term for carrying capacity, K_P, for the predator population is included. What assumption about the biology underlies this?
b. Consider a case where the assumption in part a. is no longer justified. How would you modify the Lotka-Volterra equations to account for this? (Hint: You should think of the term, $\varepsilon\alpha N$, as a per-capita population growth rate for the predator population, which is analogous to the per-capita growth rate, r, for the prey population.)
c. Find the fixed points for the new set of equations in part b.
d. Perform a stability analysis to see if this system is stable around the non-trivial (non-zero) fixed point. That is, compute the Jacobian matrix at this fixed point, compute the eigenvalues of this matrix, and use the eigenvalues to determine if the system is stable, unstable, or oscillatory.
e. In class, for the unmodified Lotka-Volterra equations, we showed that $\lambda_+ = i\sqrt{Zr}$ and $\lambda_- = -i\sqrt{Zr}$. How does your answer in part d. compare with this result? How do you interpret this?
f. Is it possible to write the interaction matrix, A, for the system of modified Lotka-Volterra equations from part b. If so, write it down? If not, explain why this system does not map perfectly onto our interaction matrix, A, from class.

4. **(30 points)** Consider a model of disease spread that contains a compartment for the density of susceptible (S) individuals, infective (I) individuals, and chronic carrier (C) individuals. The chronic carriers (C) become infected such that they do not show symptoms of the disease, remain infected forever, and will continually transmit the disease to susceptibles (S).

$$\frac{\mathrm{d}S}{dt} = -\beta IS - \alpha SC + \gamma I$$
$$\frac{\mathrm{d}I}{dt} = \beta IS - \gamma I$$
$$\frac{\mathrm{d}C}{dt} = \alpha SC$$

where β is the average rate at which infectives are infecting susceptibles, α is the average infection rate from susceptibles to chronic carriers, and γ is the recovery rate from infectives back to susceptibles.

a. Is the total density of individuals, $N = S + I + C$, constant in time or does it change in time. Use the equations above to prove your answer. Can you use your result to reduce the number of independent variables and equations?
b. Find all of the fixed points for this system
c. To see if a newly introduced disease will spread, which fixed point should you examine and perform a stability analysis around?
d. Perform a stability analysis to determine whether a new disease will spread. For this part, begin by computing the Jacobian matrix around the fixed point in part c., and then computing the eigenvalues of this Jacobian.

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e. **(required for 286 graduate students or 5 points extra credit for 186 undergraduate students)** Express your eigenvalues in terms of two ratios: $R_0 = \frac{\beta N}{\gamma}$ and $\widetilde{R_0} = \frac{(\alpha+\beta)N}{\gamma}$. What is the biological meaning of these two ratios?

f. **(required for 286 graduate students or 5 points extra credit for 186 undergraduate students)** Is the system stable or unstable when $R_0 > 1$? Is the system stable or unstable when $\widetilde{R_0}<1$? Explain why these conditions correspond to specific cases for the dynamics? That is, explain why you do not need to know the values of *both* R_0 and $\widetilde{R_0}$ for this system.

5. **(required for 286 graduate students) (10 points)** Consider a system with two eigenvalues: $\lambda_+ = -\alpha + i\sigma$ and $\lambda_- = -\alpha - i\sigma$ that describe the stability of a system around a specific fixed point, with σ being a constant real number. The value of the term α arises from a mutation after t=100 days. Before t=100 days, there is no real part to the eigenvalues because α=0. After t=100 days, α is time dependent and follows the differential equation

$$\frac{\mathrm{d}\alpha(t)}{dt} = m\alpha(t)$$

where m is a positive constant.

a. Describe the dynamics of this system around this fixed point in the case that the time t is less than100 days, t is just a little greater than 100 days, and t is much greater than 100 days.
b. If m switches signs and becomes negative, describe the dynamics of this system around this fixed point in the case that the time t is less than100 days, t is just a little greater than 100 days, and t is much greater than 100 days.

REMINDER: The Jacobian for the Lotka-Volterra system in problem 3 is defined by

$$J^* = \begin{pmatrix} \left.\partial f(N,P)\middle/\partial N\right|_{\substack{N=N^* \\ P=P^*}} & \left.\partial f(N,P)\middle/\partial P\right|_{\substack{N=N^* \\ P=P^*}} \\ \left.\partial g(N,P)\middle/\partial N\right|_{\substack{N=N^* \\ P=P^*}} & \left.\partial g(N,P)\middle/\partial P\right|_{\substack{N=N^* \\ P=P^*}} \end{pmatrix}$$

where

$$\frac{\mathrm{d}N}{dt} = rN - \alpha NP \equiv f(N,P)$$
$$\frac{\mathrm{d}P}{dt} = \varepsilon\alpha NP - ZP \equiv g(N,P)$$

The Jacobian for the *SIR* model in problem 4 is defined the similarly but starting from the *SIR* equations rather than the Lotka-Volterra equations.